THE DEFINING EQUATIONS OF A RELATIVE GLOBAL COMPLETE INTERSECTION FORM A KOSZUL REGULAR SEQUENCE

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Allowed standard prerequisites (editorial).

- The homology of the Koszul complex of a sequence is annihilated by the ideal generated by the sequence. used at proof lines 8393-8395. Reduces the verification to primes containing that ideal.
- The conormal description of a relative global complete intersection, which makes the equations a regular sequence locally. Algebra, Lemma \ref{algebra-lemma-relative-global-complete-intersection-conormal}, cited at proof lines 8398-8400. Gives a regular sequence at each relevant prime.
- A regular sequence is Koszul regular. Lemma \ref{lemma-regular-koszul-regular}, cited at proof lines 8400-8401. Converts the local regularity into the Koszul regularity that is claimed.

Source reference. Original lemma 07D2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9397–9397. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1053–1053. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 062F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7794–7794. Navigation pointer to a statement the

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Lemma 0.2. *Let R be a ring. If $R[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$ is a relative global complete intersection, then $f_1, \ldots, f_c$ is a Koszul regular sequence.*

Proof. Recall that the homology groups $H_i(K_\bullet(f_\bullet))$ are annihilated by the ideal $(f_1, \ldots, f_c)$. Hence it suffices to show that $H_i(K_\bullet(f_\bullet))_{\mathfrak{q}}$ is zero for all primes $\mathfrak{q} \subset R[x_1, \ldots, x_n]$ containing $(f_1, \ldots, f_c)$. This follows from Algebra, Lemma 00SV and the fact that a regular sequence is Koszul regular (Lemma 062F). □